

Inline and Display Mathematics

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\int_{\Omega} \left(\left\langle \frac{\partial d_{\epsilon}(t)}{\partial x_i}, \frac{\partial d_{\epsilon}(t)}{\partial x_j} \right\rangle \frac{\partial Y^i}{\partial x_j} - \left(\frac{1}{2} |\nabla d_{\epsilon}(t)|^2 + \frac{1}{4\epsilon^2} (1 - |d_{\epsilon}(t)|^2)^2 \right) \operatorname{div} Y + \left\langle \tau_{\epsilon}(t), Y \cdot \nabla d_{\epsilon}(t) \right\rangle \right) = 0.$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\left| \sum_{\alpha\beta\gamma\delta} \langle \Phi_{\alpha} \Phi_{\beta} \Phi_{\gamma} \Phi_{\delta} \rangle_c / \sum_{\alpha\beta\gamma\delta} \langle \Phi_{\alpha} \Phi_{\beta} \Phi_{\gamma} \Phi_{\delta} \rangle \right| = |\langle \Phi^4(x) \rangle_c / \langle \Phi^4(x) \rangle| \quad (n = 4) ,$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\int_0^R \mathrm{d}r \left[u_{1\epsilon\kappa}(r) u_{1\epsilon'\kappa}(r) + u_{2\epsilon\kappa}(r) u_{2\epsilon'\kappa}(r) \right] = \frac{u_{1\epsilon\kappa}(R) u_{2\epsilon'\kappa}(R) - u_{1\epsilon'\kappa}(R) u_{2\epsilon\kappa}(R)}{\epsilon - \epsilon'} .$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\bar{\mathcal{G}}_{(\mathrm{I})}^{(\rho)a} \Gamma^{(\rho)} = \Delta_{\mathrm{cl}(\mathrm{I})}^{(\rho)a} + O(\rho^2) \quad , \quad \bar{\mathcal{G}}_{(\mathrm{II})}^{(\rho)a} \Gamma^{(\rho)} = \Delta_{\mathrm{cl}(\mathrm{II})}^{(\rho)a} + O(\rho^2) \quad ,$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$Z[J^\mu, \bar{\eta}, \eta] = \int [dA_\mu] \mathfrak{S}(A_\mu) \exp \left(- \left\langle \frac{1}{4} F_{\mu\nu} F^{\mu\nu} \right\rangle + \langle J^\mu A_\mu \rangle + \left\langle \bar{\eta}' (i\gamma^\mu \partial_\mu)^{-1} \eta' \right\rangle \right) ,$$